Hagar Alaaeldien

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Professional Summary

2x ISTQB® Certified Junior QA Engineer with hands-on experience in manual, functional, integration, regression, and API testing across web, mobile, and IoT applications. Skilled in test case design, exploratory and edge-case testing, defect reporting, and SDLC/STLC practices using Jira, with a strong programming foundation in Python, Java, and SQL. Experienced in using AI tools such as ChatGPT and Claude to support test design, analysis, troubleshooting, and QA workflows.

Education

BSc. - Communication & Information Engineering Oct 2020 – Jun 2025

University of Science and Technology at Zewail City

- **Graduation Project:** Voice-controlled Assistive Soft-robotic Glove
- **Graduation Project Grade:** A

Experience

Software Testing Trainee, *Digital Egypt Pioneers Initiative* Jul 2026 – Present

- Applied ISTQB-based software testing principles throughout the Software Development Life Cycle (SDLC), covering test analysis, test design, and test management.
- Developed and executed manual test cases, documented and tracked defects using Jira
- Developed professional communication, presentation, teamwork, and problem-solving skills through technical and soft-skills training.

Business Development Representative, *P-Vita* Feb 2025 – Jul 2025

- Researched and tracked grants, incubators, and VC opportunities aligned with startup goals.
- Organized and validated opportunity requirements, documentation, and deadlines.
- Prioritized and managed applications based on strategic fit, ensuring timely submissions.

IoT & Software Engineering Intern Oct 2023 – Jun 2024

University of Science and Technology at Zewail City

- Tested and validated an IoT-enabled security robot integrating facial recognition, live streaming, and environmental sensing modules.
- Performed functional and integration testing between sensors, Raspberry Pi, and mobile application components.
- Validated real-time alert generation under varied environmental and network conditions, identifying and debugging communication issues.
- Verified Firebase data logging and debugged UART communication to optimize system reliability during continuous operation.

Projects

Voice-controlled Assistive Soft-robotic Glove (Graduation Project) Oct 2024 – Jun 2025

Tools: Flutter, Firebase, Python, Raspberry Pi, Arduino, TensorFlow Lite

- Performed functional and integration testing on a voice-controlled rehabilitation glove integrating Raspberry Pi and Arduino components.
- Validated bilingual (English/Arabic) Flutter application workflows, ensuring secure data handling and reliable API communication.
- Conducted edge-case and accuracy testing for TinyML speech recognition models, optimizing wake-word detection performance.

- Utilized AI tools such as ChatGPT and Claude to support test design, test analysis, troubleshooting, and identification of edge cases during testing.
- Debugged real-time synchronization and hardware–software communication issues to ensure system stability.
- Selected as one of only 3 winning projects (out of 44) in the Edafa VC incubation program at the university's graduation exhibition.

Chatbot for Social Robot Noor, *Tools: ROS, Google Speech API, Gemini API, Python* Feb 2025 – Jun 2025

- Validated a ROS-based bilingual chatbot through scenario-based and functional testing for intent recognition and response accuracy.
- Tested API integrations and executed edge-case testing for speech misrecognition, latency, and network interruptions.
- Used Gemini API as part of the chatbot system and evaluated behavior under different input and communication conditions.

Serverless Budget Tracker, *Tools: AWS Lambda, S3, DynamoDB, HTML, CSS, JavaScript* Nov 2024 – Jan 2025

- Developed and validated a full-stack budget tracking application with serverless backend architecture.
- Performed API and integration testing between frontend, AWS Lambda functions, and DynamoDB.
- Tested data consistency, input validation, and error-handling scenarios to ensure system reliability.

Egyptian ALPR, *Tools: Python, OpenCV, TensorFlow, NumPy* Mar 2024 – Jun 2024

- Validated a real-time Arabic license plate recognition system through functional, accuracy, and dataset-based testing.
- Executed edge-case testing under varied lighting, occlusion, and low-resolution conditions, analyzing detection reliability and false positives.

ClubHub, *Tools: C#, SQL Server* Feb 2022 – Jun 2022

- Developed and tested a desktop application for managing university clubs, events, and member tasks.
- Performed CRUD validation and database integrity testing using SQL Server.
- Tested input validation, role-based access control, and user workflow scenarios.

Certifications

ISTQB® Certified Tester – Testing with Generative AI (CT-GenAI) — Jul. 2026 Credential ID: 260707007

ISTQB® Certified Tester – Foundation Level (CTFL) v4 — Jan. 2026 Credential ID: 260122016

Technical Skills

- **Testing & QA:** SDLC, STLC, Functional Testing, Integration Testing, System Testing, Regression Testing, Black-box & White-box Testing, Boundary Value Analysis, Equivalence Partitioning, Test Case Design, Test Plan Writing, Test Reporting, Defect Lifecycle, Bug Reporting, API Testing, Agile Methodologies (Scrum, Kanban, XP)
- **Tools:** Jira, Trello, Postman, SQL Server, Git, Docker, AWS (Lambda, S3, DynamoDB), Wireshark, VS Code
- **Programming:** Python, Java, C++, C, SQL
- **Platforms & Technologies:** Linux, Firebase, Flutter, Raspberry Pi, Arduino
- **AI Tools:** ChatGPT, Claude, Gemini